

Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

AIM:

To write and simulate an Embedded C program using serial interrupt in 8051 microcontroller such that while a message is continuously transmitted, any key pressed from the keyboard generates an interrupt and is displayed through UART.

APPARATUS / SOFTWARE REQUIRED:

- PC with Windows OS
- Keil μ Vision IDE
- 8051 Simulator (Keil built-in)

- Serial Window (UART #1)

PROGRAM:

```
#include <reg51.h>
```

```
#include <stdio.h>
```

```
void serial_ISR(void) interrupt 4
```

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```
char ch;
```

if (RI)

$$\{$$

```
RI = 0; // Clear receive interrupt flag ch =
```

```
SBUF; // Read received character
```

```
printf("\n>>> INTERRUPT RECEIVED: ");
```

```
printf("%c\n", ch);
```

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```

void delay(void)
{

unsigned int i;
for(i = 0; i < 50000; i++); // Delay for visibility }


void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
    TMOD = 0x20; // Timer1 mode 2 (auto-reload) TH1 =
    0xFD; // Baud rate 9600
    TR1 = 1; // Start Timer1

    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt) TI

    = 1; // Enable printf

    while(1)
    {
        printf("Hello World\n");
        delay();
    }
}

```

OUTPUT:

